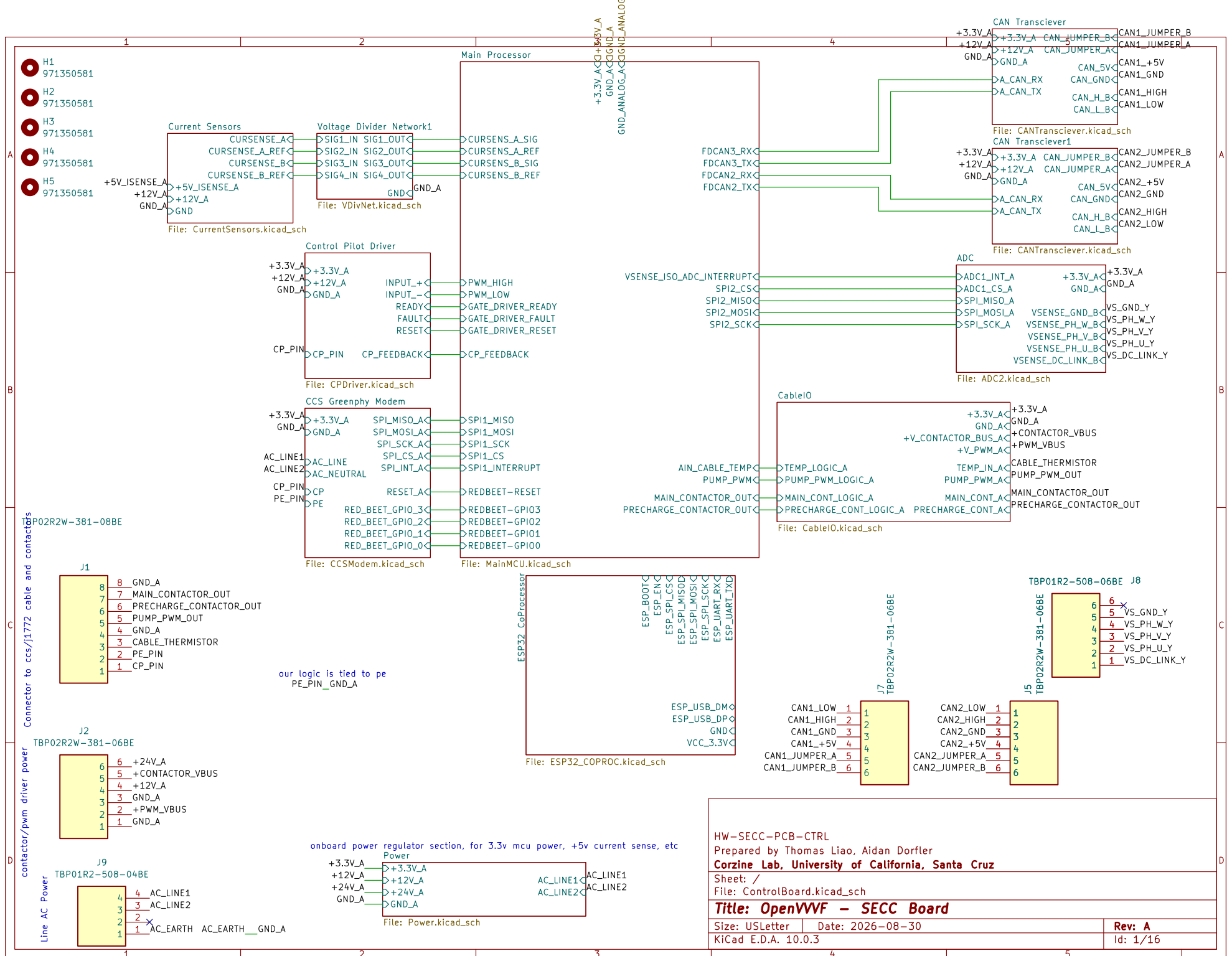
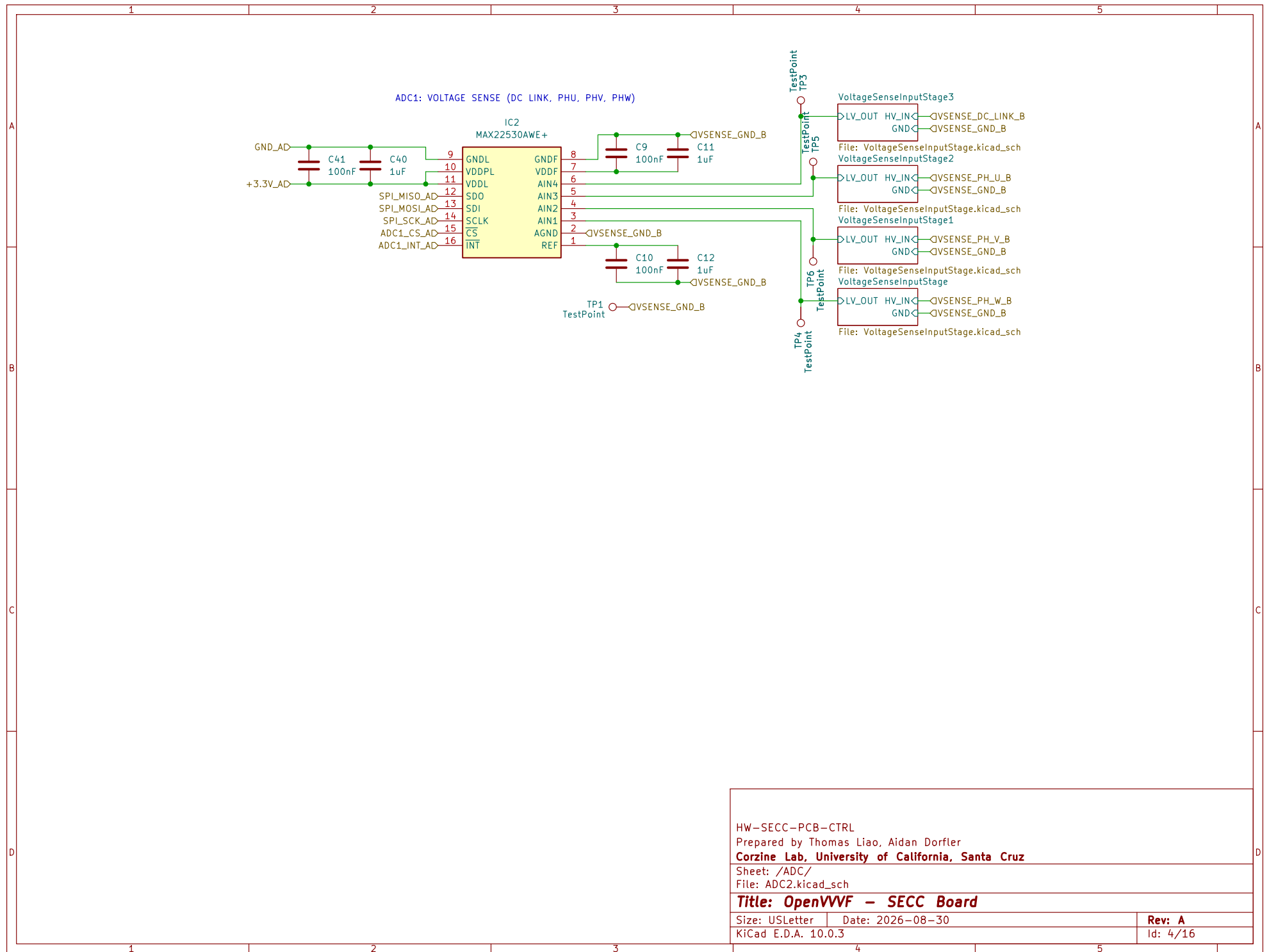






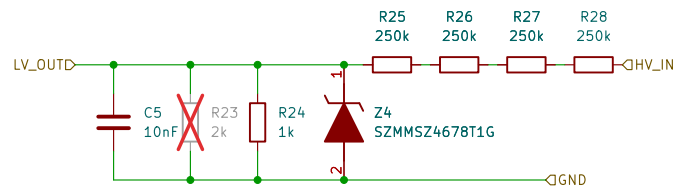
HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/
File: ADC2.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30
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Rev: A
Id: 4/16



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-SECC-PCB-CTRL

Prepared by Thomas Liao, Aidan Dorfler

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/

File: VoltageSenseInputStage.kicad_sch

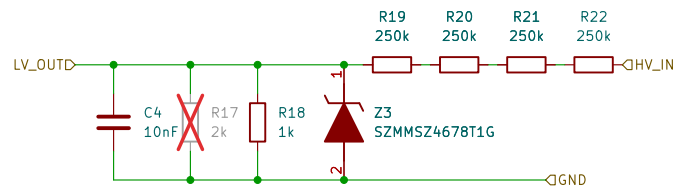
Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30

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Rev: A

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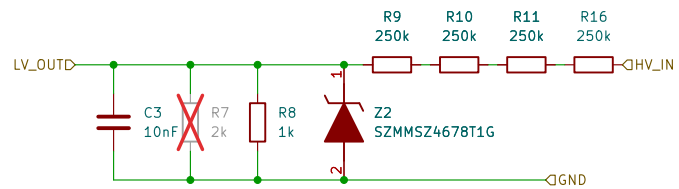
ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30
KiCad E.D.A. 10.0.3

Rev: A
Id: 3/16



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-SECC-PCB-CTRL

Prepared by Thomas Liao, Aidan Dorfler

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

Title: OpenVWF - SECC Board

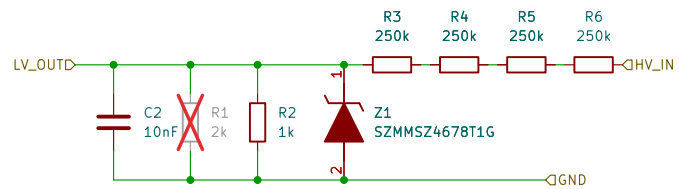
Size: USLetter

Date: 2026-08-30

Rev: A

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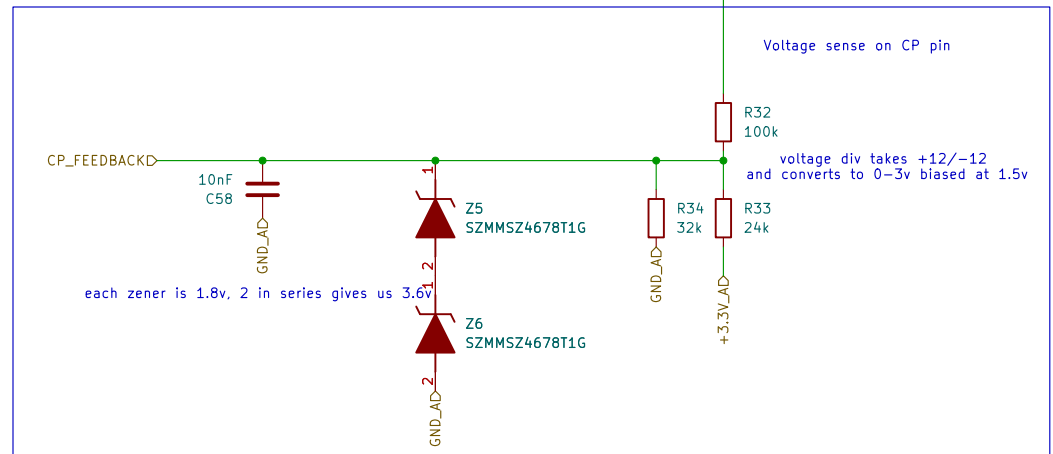
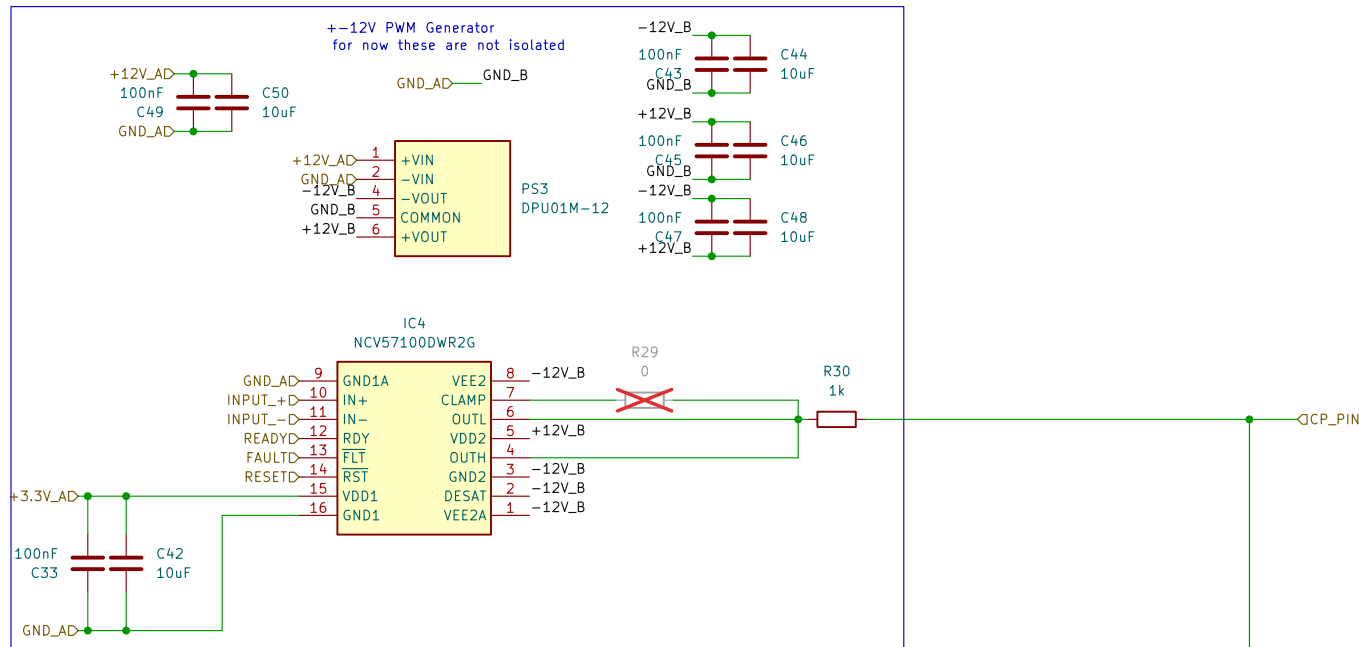
ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30
KiCad E.D.A. 10.0.3

Rev: A
Id: 6/16



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Prepared by Thomas Liao, Aidan Dorfler

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Sheet: /Control Pilot Driver/

File: CPDriver.kicad_sch

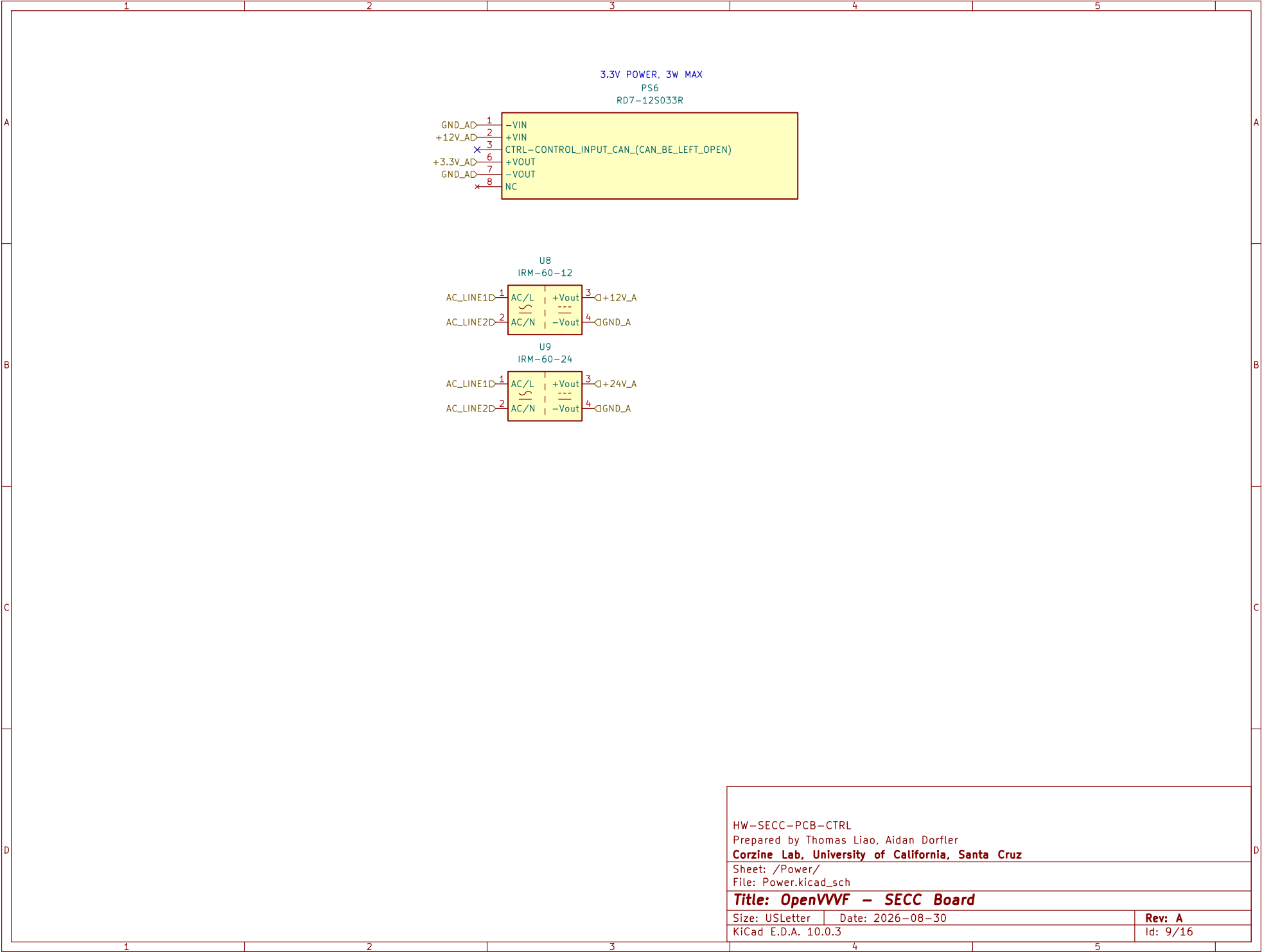
Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30

KiCad E.D.A. 10.0.3

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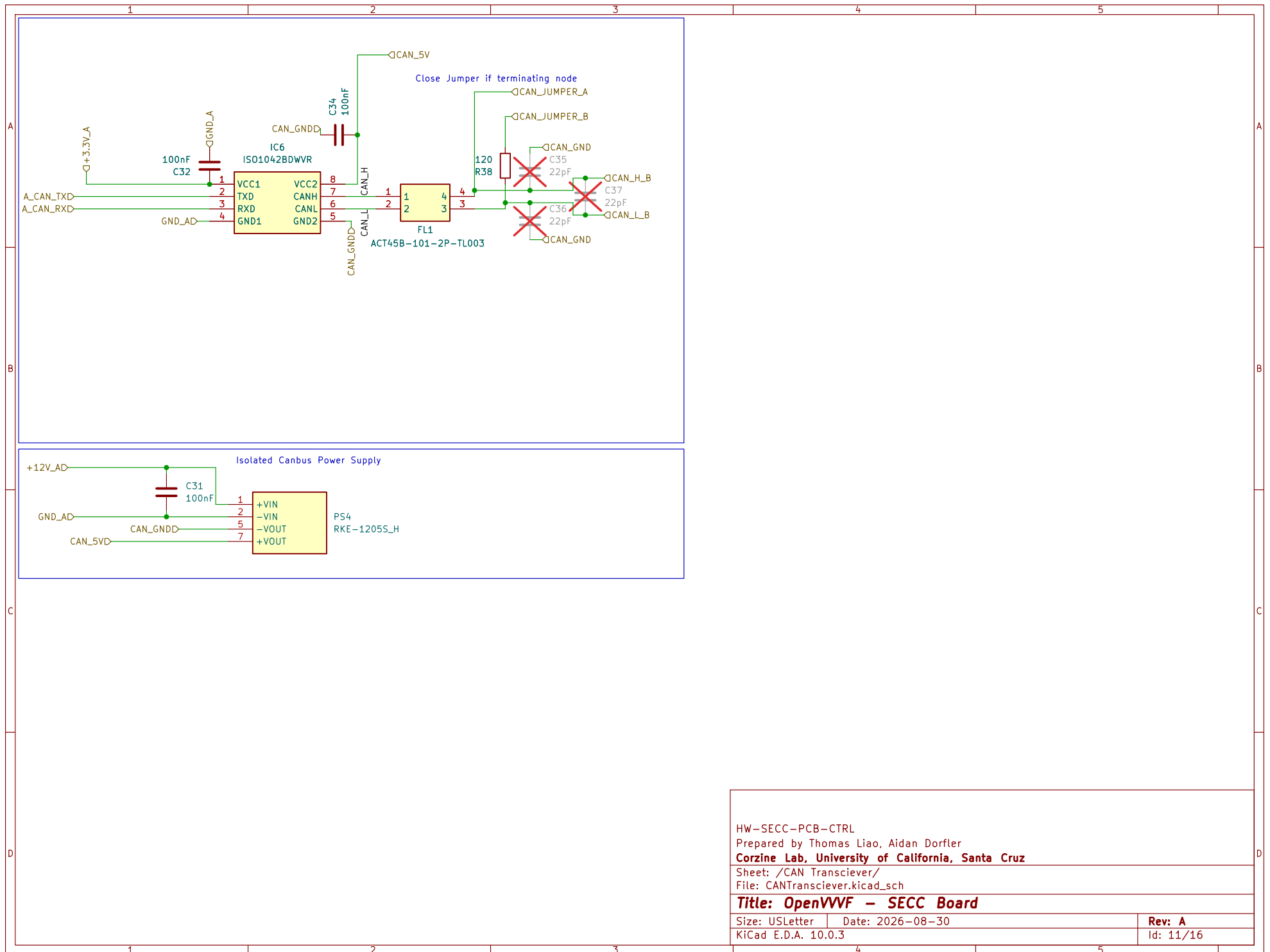


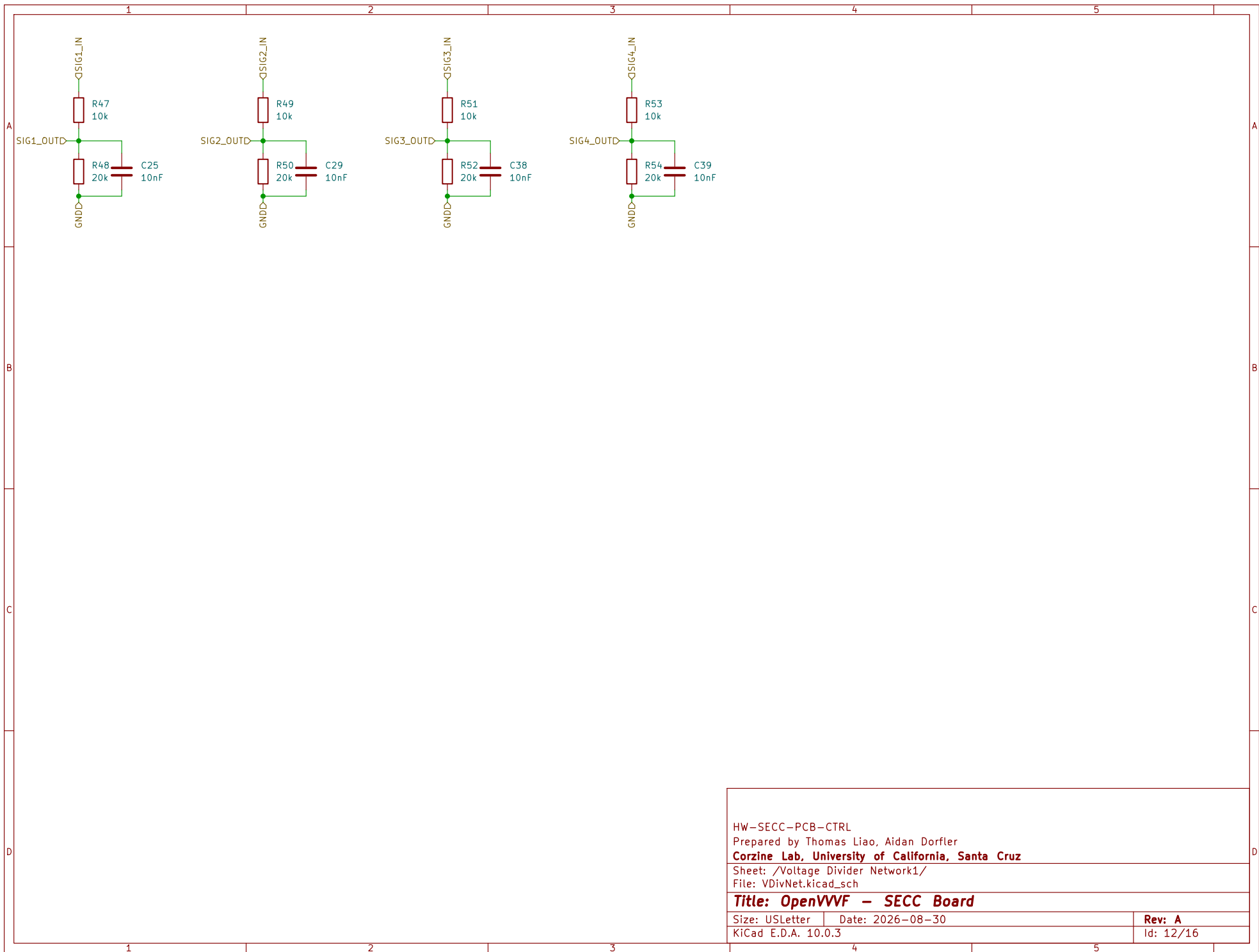
HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz

Sheet: /Power/
File: Power.kicad_sch

Title: OpenVWF - SECC Board

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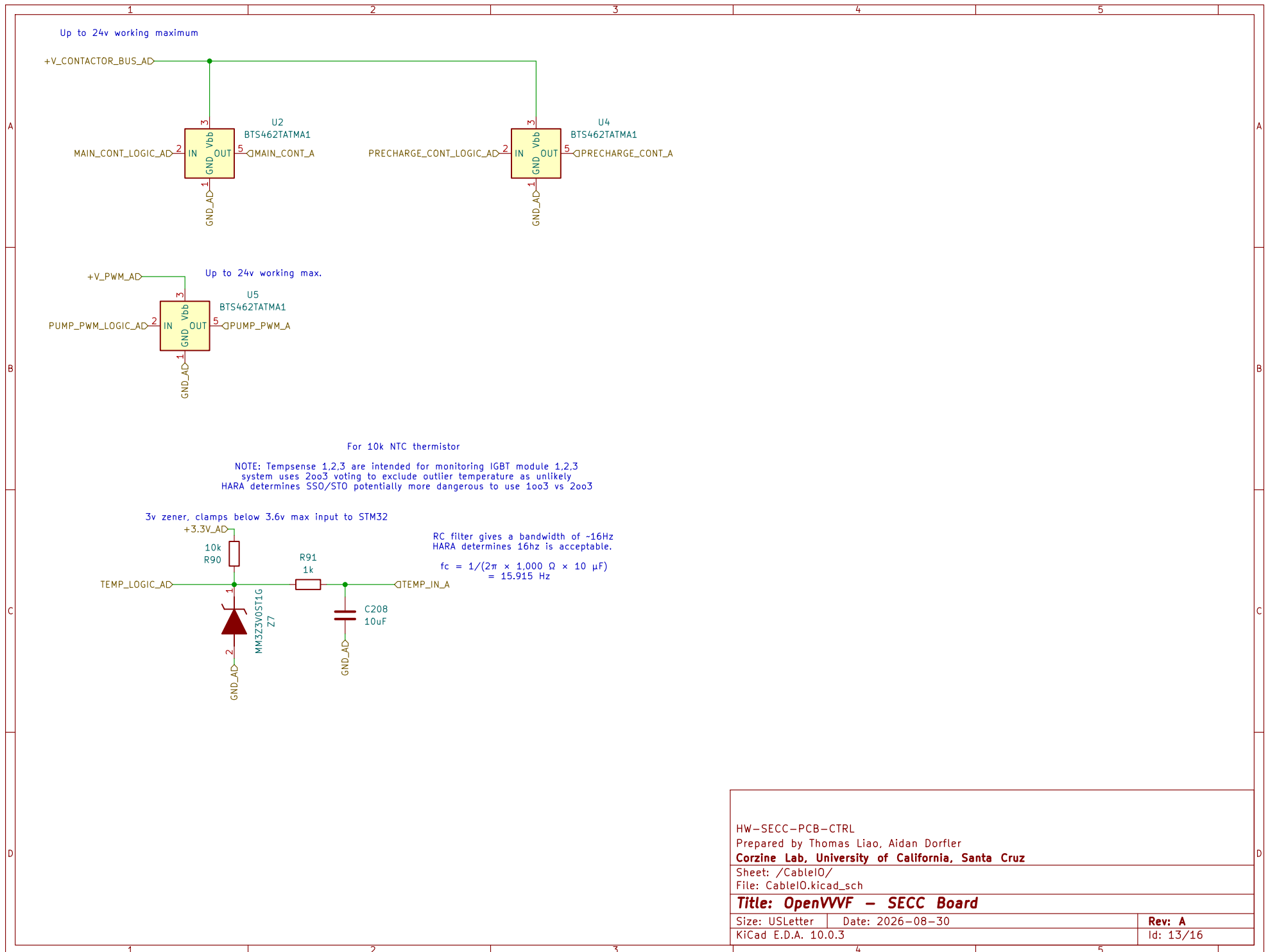




HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter	Date: 2026-08-30	Rev: A
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HW-SECC-PCB-CTRL

Prepared by Thomas Liao, Aidan Dorfler

Corzine Lab, University of California, Santa Cruz

Sheet: /CableIO/

File: CableIO.kicad_sch

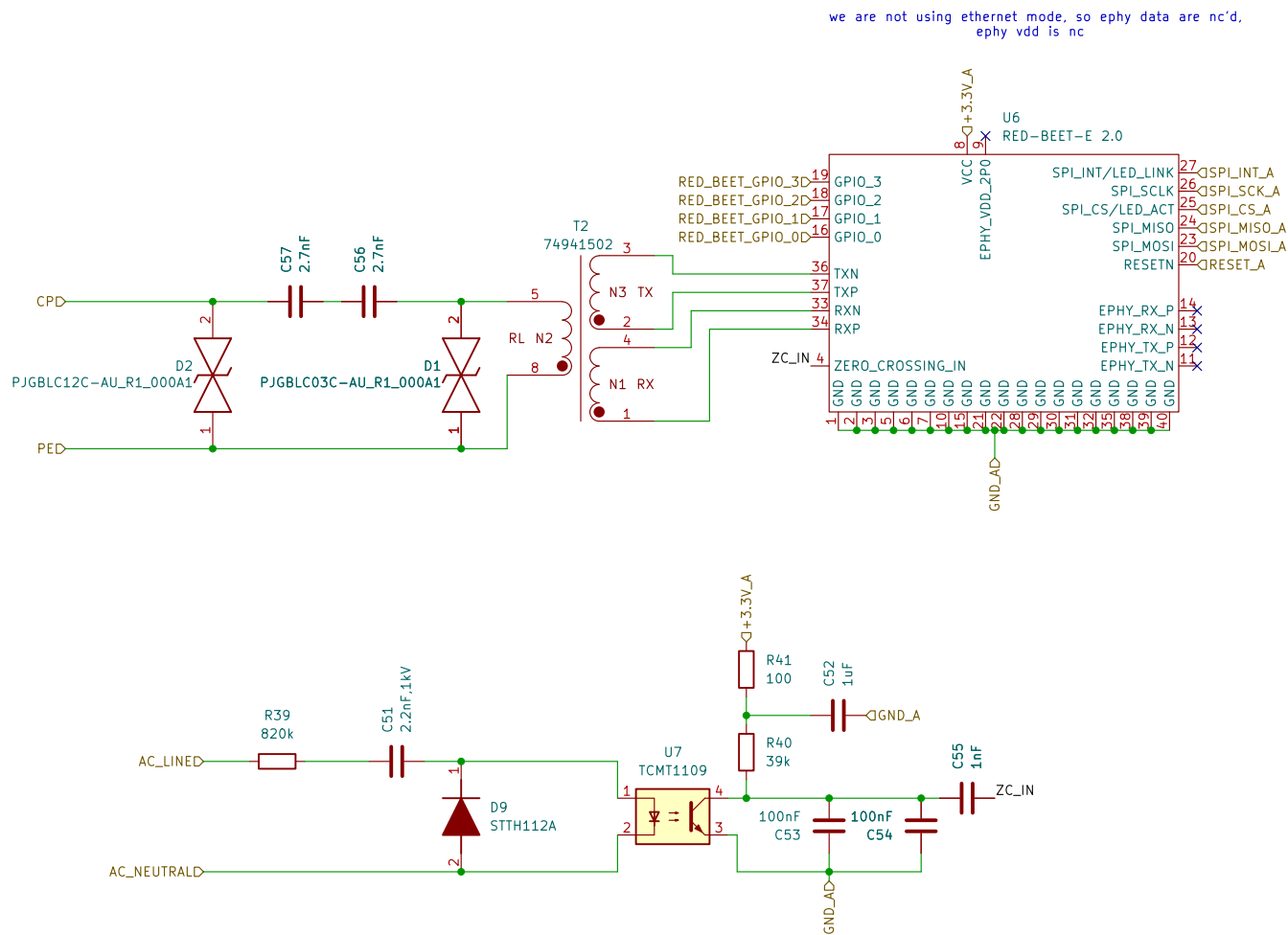
Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30

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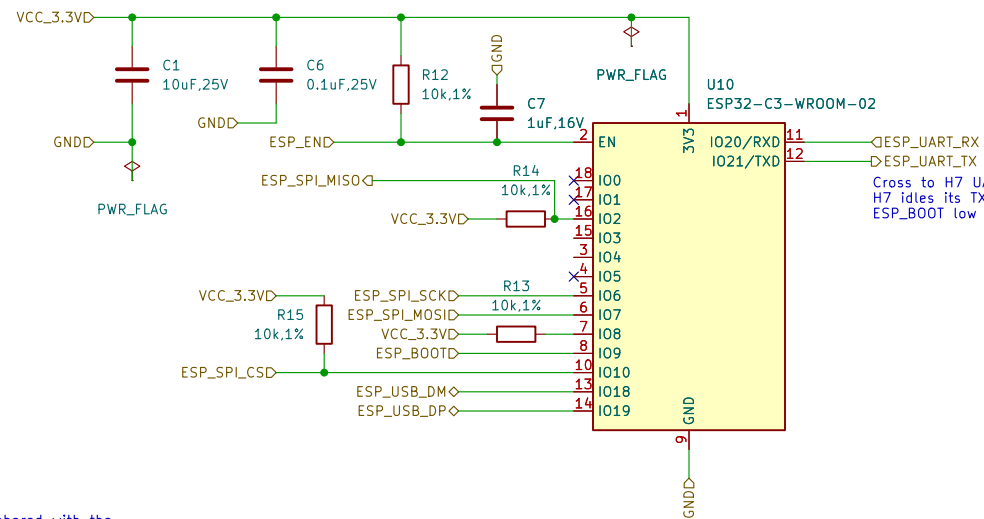
HW-SECC-PCB-CTRL
Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz
Sheet: /CCS Greenphy Modem/
File: CCSModem.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30
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3V3 rail must supply ≥ 500 mA peaks (Wi-Fi TX). Place C6 closer to pin 1 than C1.
 SPI is ESP32-C3 SPI2 in slave mode, up to 60 MHz. Keep traces short, length-matched.
 GPIO11-17 not available (in-package flash).



Cross to H7 UART: ESP_UART_TX -> H7_RX, ESP_UART_RX -> H7_TX.
 H7 idles its TX (ESP_UART_RX) high. Enter download mode:
 ESP_BOOT low + pulse ESP_EN low.

H7 drives open-drain or push-pull-LOW only. EN is shared with the R12/C7 power-on-reset RC, so the H7 must never source current into it (a push-pull high would override the RC delay and fight any pull-down on the net).

Reserved for future USB connector
 (native USB-Serial/JTAG flashing/debug).

HW-SECC-PCB-CTRL
 Prepared by Thomas Liao, Aidan Dorfler
Corzine Lab, University of California, Santa Cruz

Sheet: /ESP32 CoProcessor/
 File: ESP32_COPROC.kicad_sch

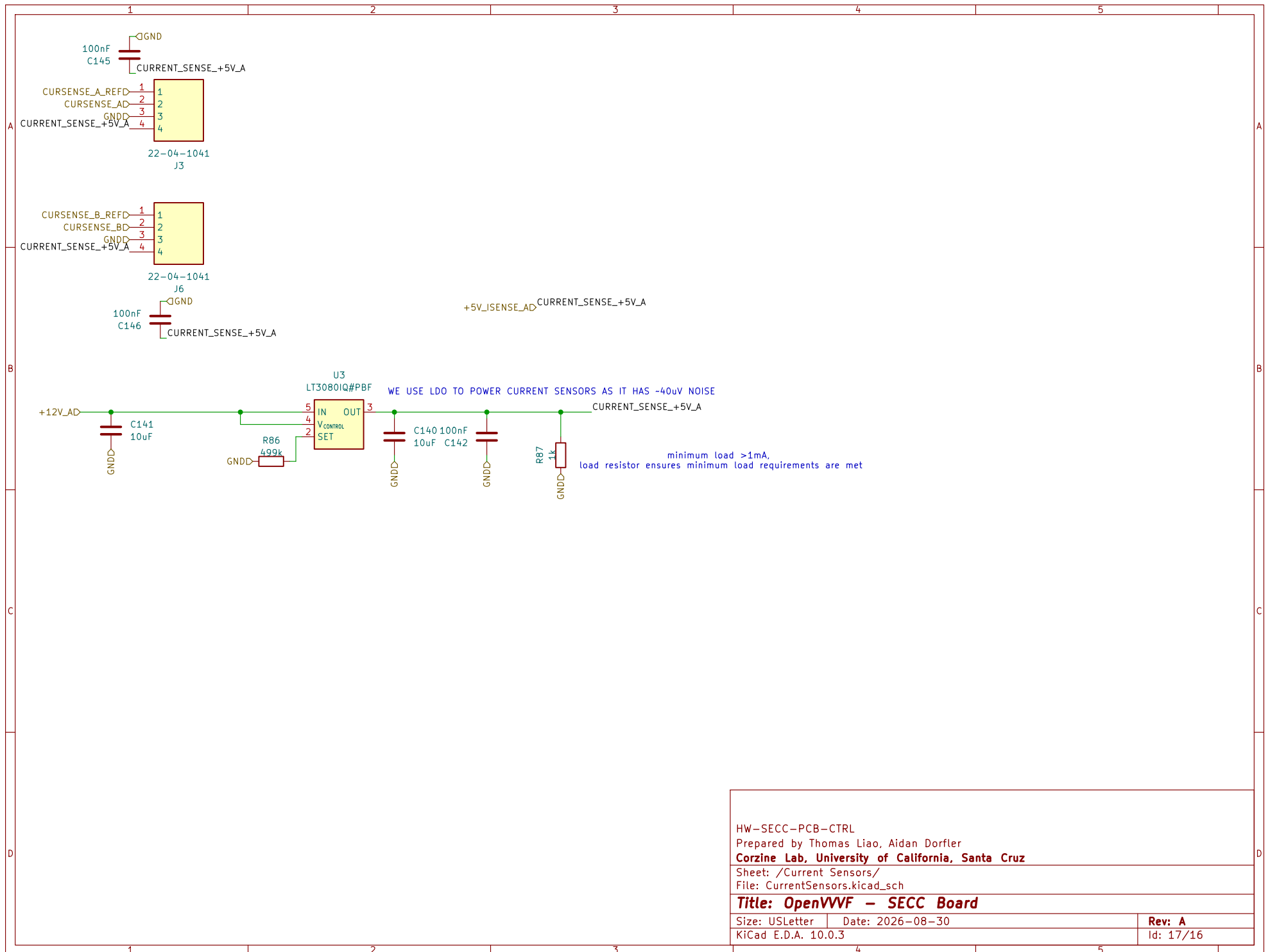
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Prepared by Thomas Liao, Aidan Dorfler

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Sheet: /Current Sensors/

File: CurrentSensors.kicad_sch

Title: OpenVWF - SECC Board

Size: USLetter Date: 2026-08-30

KiCad E.D.A. 10.0.3

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